Heat recovery in a small bakery

A bakery oven releases a large amount of warm air through its chimney. We fitted a small heat exchanger that captures part of this heat for the wash water. The exchanger has a primary circuit on the chimney side and a secondary circuit on the water side. This note documents the layout and the first month of measurements.

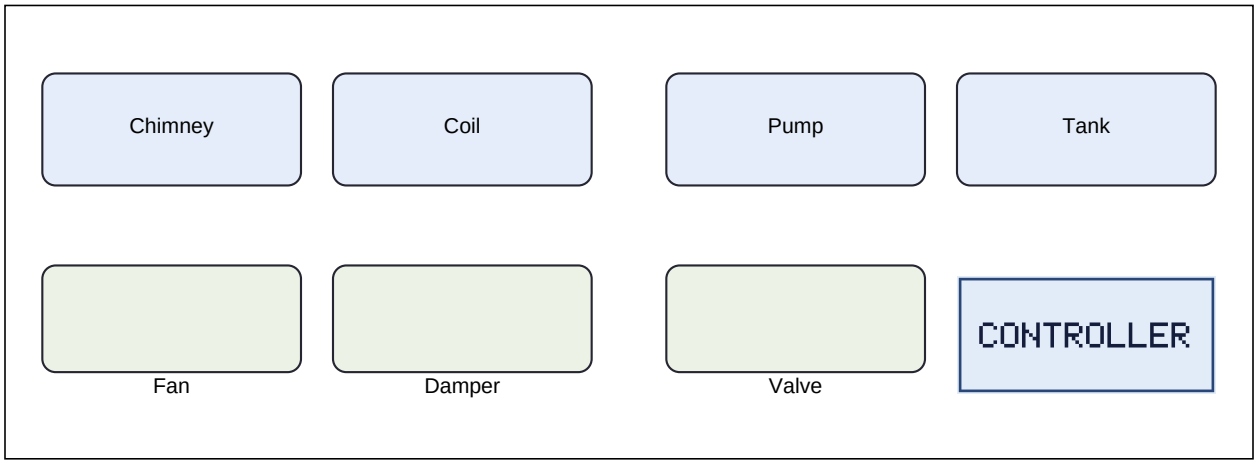

Figure 1: Heat exchanger layout. The left half shows the primary circuit and the right half shows the secondary circuit.

Hot air leaves the oven and passes a coil before it reaches the chimney. A fan keeps the air moving when the oven door is open and the natural draught is weak. On the water side, a pump circulates the wash water between the coil and an insulated tank. A small controller switches the pump on when the coil is warmer than the tank.

The controller is an old unit with a printed front panel and no digital interface. We read its state from the pump current instead of from the unit itself.

First month

Over the first month the exchanger heated the wash water by about eighteen degrees on average. Gas use for water heating fell by roughly a third compared with the same month last year. The bakers reported no change in oven behaviour and no extra cleaning work.

Table 1: Weekly averages during the first month.

Week	Water in	Water out	Gas saved
1	12 C	29 C	28 %
2	11 C	30 C	33 %
3	12 C	31 C	35 %
4	13 C	31 C	34 %

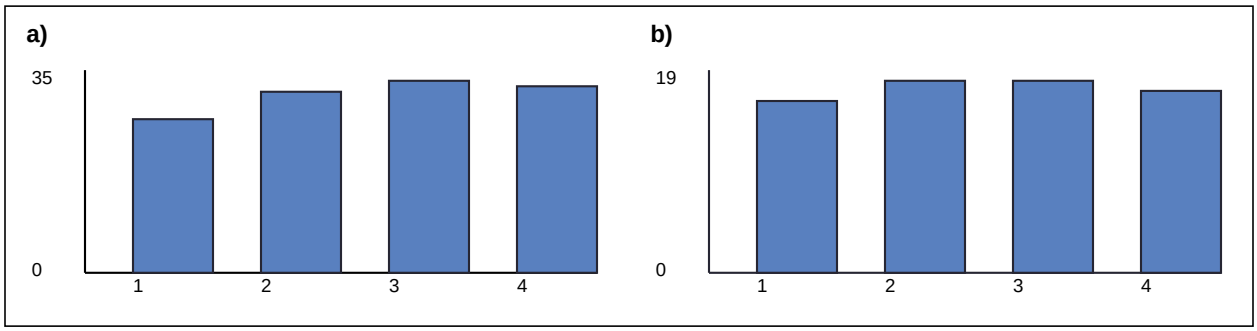

Figure 2: Weekly results. Panel a shows gas savings and panel b shows the water temperature rise.

Savings grew in the second week after we cleaned soot from the coil. We expect a small drop in summer, when the incoming water is already warmer. A full year of data will tell whether the exchanger pays for itself within three years.